

Teaching Team:

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GENETICS

BIOL 3023 | SPRING 2026

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THERE IS NO KNOWLEDGE THAT IS NOT POWER

Ralph Waldo Emerson

Course Overview

Every life science discipline and healthcare field depends on thorough familiarity with genetics principles to be intelligently applied and every life on earth is influenced by these disciplines and fields. Students in this course will solidify their understanding of the the central dogma of genetics, broaden their grasp on genomic regulation, and apply basic genetic principles learned in high school and earlier college courses to developing a more comprehensive understanding of molecular genetics, heredity, and population genetics.

Where to Find

Canvas Modules – Where you will find

instructional materials and turn in ALL

assignments & quizzes organized by week

Microsoft Teams – Where you communicate

with the leadership team and with classmates

What to Do

Every week you will have

Asynchronous (complete at your own pace)

- A set of instructional materials to complete (videos and readings)
- A homework assignment

Synchronous, in person

- A required meeting with your small group
- An in-person quiz

How does this course work?

Canvas -> Modules -> Weekly Tasks

- Each week a new page will appear in this module. It will be designated by the Week & Date (Week 1 Sept 12).
- ***All*** learning materials and assignments for that week will be listed on this page in the order they should be completed.
- A typical week will entail ~ 3 hours of asynchronous instructional video in ~ 20-minute blocks by topic, readings on each topic from our digital textbooks and/or the primary literature, and occasionally instruction from other resources such as podcasts.

Required Materials

Internet Access

Readings will come from freely available on-line textbooks, content created by Dr. Wiggins, peer reviewed articles, and websites created and maintained by scientific organizations (See Appendix)

Expected Behavior and Course Philosophy

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Expected Behavior

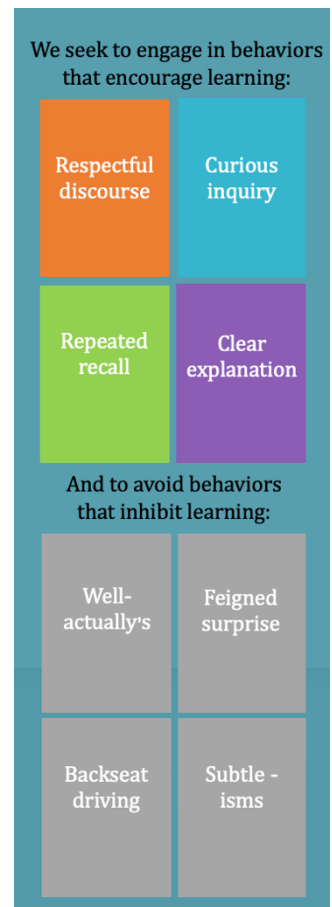
We are committed to creating a *safe and respectful space for learning*, teaching and contributing.

Instructors and students in Biol 3023 are expected to show respect and courtesy to others. All interactions should be courteous and professional regardless of platform, either online or in-person. Social learning is a central part of the course philosophy.

Therefore, in this course you are required to familiarize yourself with the social expectations that help in removing obstacles for learning. These are the [four social rules](#) by the [Recurse Center](#). These rules are in addition to the expectations outlined in the [OSU Student Code of Conduct](#).

Rudeness to your peers, TA, or instructor in any forum will not be tolerated.

You are further expected to communicate with the Instructional Team regarding questions, technical difficulties, conflicts, etc.



Course philosophy (detailed post [here](#))

Why This Course Uses Tactile and Active Learning

Research in cognitive science and education consistently shows that students learn more deeply when they actively engage with material rather than passively receive it. Activities such as drawing, physically manipulating models, moving through problems, and explaining ideas aloud strengthen memory formation and conceptual understanding by engaging multiple neural pathways. Although students often *feel* that they are learning less during these active, hands-on sessions compared to listening to lectures, studies show the opposite is true: comprehension, retention, and ability to transfer knowledge are significantly higher. The discomfort associated with active learning reflects productive cognitive effort, not ineffective teaching, and is a normal and important part of the learning process.

For this reason, a substantial portion of class time is spent in small-group, tactile, and kinesthetic activities. These sessions are **not graded**. Their purpose is to create a low-stakes environment where you can test ideas, ask questions, and practice reasoning without penalty. Making mistakes during these activities is not only allowed but expected—mistakes are a critical part of how the brain learns. Identifying, discussing, and correcting those mistakes is what leads to durable understanding.

Learning From Mistakes: Homework Dissections (See Appendix: Homework)

Homework in this course is intentionally challenging and designed to push you beyond memorization. To support learning from this process, you will have the opportunity to complete **homework dissections**. For each homework set, you may select **one homework question** to dissect in depth.

In a homework dissection, you will analyze:

- what your original reasoning was,
- where and why it broke down,
- what the correct reasoning or approach should have been, and
- how your understanding has changed as a result.

The goal of a homework dissection is not simply to fix an answer, but to practice *metacognition*—thinking about how you think.

Research shows that this kind of structured reflection and correction is one of the most effective ways to solidify learning and prevent the same mistakes in the future.

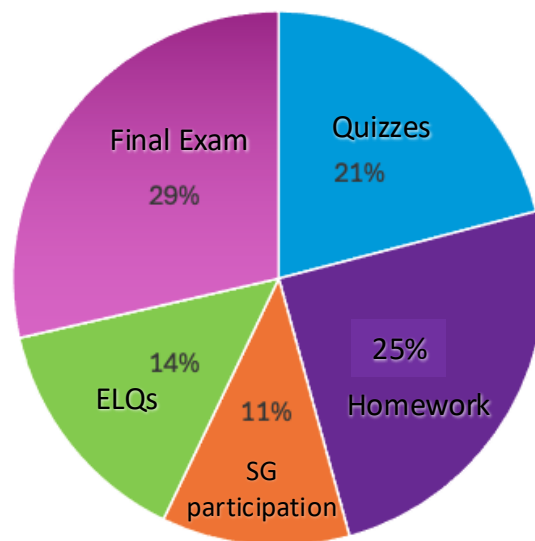


HOW LEARNING IS ASSESSED

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Homework: Every week you will have a homework assignment that reinforces the concepts covered that week and integrates concepts from the previous weeks. The homework will be due the following Thursday by 11:59pm.

There are 14 homework assignments, each accounts for 10 points. Each student's lowest homework grade will be dropped. Homework accounts for a total of 130 course points.



Every single word of your homework must be written in your own voice. Every diagram or drawing must be entirely your own work. Any answer that contains material not written in your own voice will earn zero points. *See page 5 for the course AI acceptable use statement*

You will have the opportunity to dissect one question from each homework assignment for up to half credit. For details see “homework” in the appendix.

Weekly quizzes: Every week during the first 10 minutes of small group you will take a short, 5-6 question quiz. The questions will consistently be higher-order thinking, synthesis, and integration questions. There will be 13 quizzes. The lowest two (2) quiz grades will be dropped. Each quiz is worth 10 points. Quizzes account for 110 total course points. *See accessibility on page 4 for notes about SAS and the quizzes.

Embedded learning questions (ELQs): Each instructional video contains embedded learning questions. These are designed to highlight important points and ensure you are engaged and *can be retaken an unlimited number of times before the due date*. Embedded learning questions are worth a total of ~ 5 points each week.

***Important note:** just because a concept does not have an accompanying embedded learning question this does not mean it is unimportant or will not be tested.

Small group meetings: Small group meetings are synchronous and in person. You must attend at *your scheduled meeting time*. Attendance and participation in small group meetings is worth 5 points per week. Your lowest two (2) small group participation points will be dropped. Small group participation accounts for 60 total course points.

Final Exam: The final exam is comprehensive, in-person, on paper. It will be held during finals week at a TBD day and time (the unusual structure of the course makes scheduling a little tricky – I will let you know as soon as I do). The final exam is worth 150 total course points.

An important note on ‘drops’: The two lowest quiz and small group grades will be automatically dropped. This policy is in place to accommodate the myriad ways life happens (illness, travel, etc.). If you have to miss small group there isn’t a make-up for the participation points or for the quiz. The zero will be dropped.

A : $\geq 89.5\%$ B : $\geq 79.5\%$ C : $\geq 69.5\%$ D : $\geq 59.5\%$

Grades will not be bumped, fudged, rounded etc. Take advantage of the embedded question re-takes and the homework dissections. If email to ask that your letter grade be changed you will not receive a reply.



Basic needs: Any student who has difficulty affording groceries or accessing sufficient food to eat every day, or who lacks a safe and stable place to live is urged to contact the [office of the Vice President for Student Affairs](#) (vpsao@okstate.edu) for support. Furthermore, please notify me if you are comfortable in doing so. This will enable me to provide any resources that I can.

Accessibility: If you have a disability (e.g., learning disability, temporary or permanent physical handicap) or even suspect you might - that requires accessibility services (e.g., note-taking, particular testing needs), please see Student Accessibility Services (accessibility.okstate.edu). They can instruct you regarding accommodations. It is then your responsibility to inform me of your needs, if you so desire. **Please do this as early as possible** so that I can make the necessary arrangements. I am legally bound to treat such information as confidential. I will be happy to work with you.

*If you have testing accessibility accommodations that you plan to use for the weekly quizzes: set up your weekly quiz times *for each quiz now*, email Dr. Wiggins your testing center schedule, take each quiz at the testing center, arrive to your small group ten minutes after the official start time (when the quiz is over).

Office (student) hours: Mondays 11:30 – 12:30, Thursdays 1:30 – 2:30

Location: Life Sciences West (LSW) 211A

Communicate with me: The most important thing you can do and the theme running through this entire syllabus is **TALK TO ME**. I can't help if I don't know you need it. Please don't be like hat-fire-fine-dog.



WHO TO CONTACT

If at any point you to discuss any aspect of this (or another) course and do not feel comfortable coming to me (or the instructor of another course) you have a number of options. Below I've listed who you can contact about this course.

Integrative Biology Department Head: Dr. Jason Belden (my direct supervisor) jbelden@okstate.edu

[Page](#) where you can find the College of Arts and Sciences Leadership Team

Office of the Vice President of Academic Affairs: vpsao@okstate.edu

Vice Provost of Academic Affairs: Dr. Jeanette Mendez jeanette.mendez@okstate.edu



AI Acceptable Use Statement

Artificial intelligence (AI) tools (e.g., ChatGPT, Gemini, Claude, Copilot etc.) may be used **only as a learning aid** to support your understanding of genetics concepts. All homework must be completed **independently and in your own voice**, as described below.

Permitted Uses

You **may** use AI tools to:

- Explore or review **general genetics concepts** covered in lecture, readings, or prior coursework.
- Ask for **alternative explanations, analogies, or summaries** of topics (e.g., “Explain genetic recombination in simple terms”).
- Identify areas of confusion to guide further studying.
- Help plan how to study or how to approach learning the material (not how to answer a specific question).

Prohibited Uses

You **may not** use AI tools to:

- Answer, solve, or complete **any homework question**, in whole or in part.
- Paste, upload, paraphrase, or otherwise submit **any homework prompt, question, dataset, figure, or scenario** into an AI tool.
- Use AI to generate explanations, calculations, interpretations, or written responses that directly address a homework question.
- Use AI-generated language, reasoning, or structure as a template for your answers, even if you rewrite or modify it.

“In Your Own Voice” Requirement

All homework responses must be written **in your own voice**, reflecting your personal understanding of the material. This means:

- You must be able to explain and defend your reasoning without reliance on external tools.
- Your writing should be consistent with your prior work, course level, and instructional content.
- Work that reflects AI-style phrasing, structure, or reasoning—rather than your own thinking—may be treated as unauthorized assistance.

AI may help you learn the material, but it *may not* help you produce the work.

Academic Integrity

Use of AI beyond the limits described above constitutes **unauthorized assistance** and a violation of the course’s academic integrity policy. This includes cases where AI use cannot be conclusively proven but the submitted work shows strong evidence of external generation or assistance.

If you are unsure whether a specific use of AI is allowed, **do not use it** and ask the instructor *before* completing the assignment.

Why This Policy Exists: Learning genetics requires you to **practice thinking through problems yourself**. Homework is so much less about getting the correct answer than it is about developing your ability to reason, explain biological processes, and apply concepts in your own words.

Homework questions are thoughtfully designed to walk you through the thought processes to understand the concepts.

If an AI tool does the thinking, structuring, or explaining for you, then *you* are not doing the learning. This policy exists to ensure that homework supports real understanding, prepares you for exams and future courses, and reflects your own developing scientific voice.



Approximate Course Schedule

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Week	Monday	Content Covered
1	12-Jan	Course Intro/What is a Gene/How is DNA organized
2	19-Jan	Chromosomal organization/Ploidy
3	26-Jan	Transcription
4	2-Feb	Transcriptional Regulation/Epigenetics
5	9-Feb	mRNA/Translation
6	16-Feb	Translational control/Mutation
7	23-Feb	Mutations cont/Cell cycle / replication
8	2-Mar	Replicaton cont. / Mitosis / Meiosis
9	9-Mar	Meiosis cont. / Genetic recombination
10	16-Mar	Spring Break
11	23-Mar	Independent assortment and linkage / Polygenic traits
12	30-Mar	Mendelian genetics / Sex determination
13	6-Apr	Population genetics
14	13-Apr	Misuse of genetics
15	20-Apr	Genomic tools
16	27-Apr	Genomics in research and medicine
Finals Week	4-May	In-person finals by small group section

Assignments and Due Dates

	Monday	Quizzes (Taken in Small Group)	Homework due (Thursday 11:59)	Small group meeting	ELQs due (Sunday 11:59)
Week 1	12-Jan	No Quiz	No HW	1	Week 1
Week 2	19-Jan	No meetings	1	No Meetings	Week 2
Week 3	26-Jan	1	2	2	Week 3
Week 4	2-Feb	2	3	3	Week 4
Week 5	9-Feb	3	4	4	Week 5
Week 6	16-Feb	4	5	5	Week 6
Week 7	23-Feb	5	6	6	Week 7
Week 8	2-Mar	6	7	7	Week 8
Week 9	9-Mar	7	8	8	Week 9
Week 10	16-Mar	Spring Break	Spring Break	Spring Break	Spring Break
Week 11	23-Mar	8	9	9	Week 10
Week 12	30-Mar	9	10	10	Week 11
Week 13	6-Apr	10	11	11	Week 12
Week 14	13-Apr	11	12	12	Week 13
Week 15	20-Apr	12	13	13	Week 14
Week 16	27-Apr	13	14	14	Week 15
Finals Week	Finals Taken in Person with Small Group Sections				



Textbooks (below)

Homework Dissection pg 8

Homework Dissection Example pg 9

AI use FAQ pg 10

Textbooks

Online Open Genetics:

[https://bio.libretexts.org/Bookshelves/Genetics/Book%3A_Online_Open_Genetics_\(Nickle_and_Barrette-Ng\)](https://bio.libretexts.org/Bookshelves/Genetics/Book%3A_Online_Open_Genetics_(Nickle_and_Barrette-Ng))

The PDF of the above material:

[Open Genetics, by Michael Deyholos](#). This freely downloadable text (If the first link doesn't work for you, try [this alternative link](#) instead.)

[OpenStax Biology](#). This is a big first year college textbook for students who are majoring in Biology. Its coverage of many genetics topics is quite detailed, and we will use it as another source of recommended readings. You can download the full pdf (~1500 pages) from their web site (get the low-resolution file). Note that the page number references in the Module Info handouts refer to the pdf version, not the web version.

Locke, John, 2017, "'[Open Genetics Lectures](#)' textbook for an Introduction to Molecular Genetics and Heredity (BIOL207)", <https://doi.org/10.7939/DVN/XMUPO6>, UAL Dataverse, V1; OpenGeneticsLectures_Fall2017.pdf [fileName]

Optional Physical Textbook: Modern Genetic Analysis: Integrating Genes and Genomes (2nd Ed.) Griffiths, Lewontin, Gilbert, & Miller.

- This book is older but has all of the core information and can be purchased cheaply, used.

Readings from the peer-reviewed literature as assigned (these will always be linked in the weekly Objectives and Tasks).

Additional readings from relevant sources as assigned (these will always be linked in the weekly Objectives and Tasks).



Upon receiving your graded homework you may select one question that was marked incorrect or incomplete to *dissect* for up to half question credit.

Your dissected question must

1. Be submitted by 11:59 pm on the Thursday the next HW is due
2. Include a screenshot of the graded question, including
 - The entirety of the question (even if you are only dissecting part)
 - your initial answer
 - Points earned (score)
 - any grader comments/notations.
3. Include the correct answer in your own words (the key will have been posted at this point)
4. Explain clearly and in detail what about your answer was incorrect or incomplete
5. Explain clearly and in detail what about the correct answer is different than your initial answer and
6. Explain clearly and in detail how it correctly or more fully answers the question.

The submission of a dissected answer does not guarantee points earned.

Dissections submitted without **all of** the above elements **will not be graded.**

Questions not answered cannot be dissected

See an example on the following page



Example (Imaginary) Homework

1. Question text

Thorough and well thought out answer written in your own voice that does not ramble in hopes it is 'in there somewhere' but provides a complete and comprehensive explanation/description

a. Follow up question

Answer that perhaps does not fully answer the question or is incorrect

b. Follow up question

Answer that perhaps does not fully answer the question or is incorrect

Student Homework dissection submission example

Question dissection of #1 from Homework 2

Clear indication of which question and which homework are being dissected

10/10

1. Question text

Thorough and well thought out answer written in your own voice that does not ramble in hopes it is 'in there somewhere' but provides a complete and comprehensive explanation/description

5/10

a. Follow up question

Answer that perhaps does not fully answer the question or is incorrect

b. Follow up question

2.5/10

Answer that perhaps does not fully answer the question or is incorrect

The correct answer to this question is

My answer above was incorrect/incomplete because it does/does not...

The correct answer is different from my initial answer in the following ways...

The correct answer more fully or correctly answers the question because...

Grader
comments
(if any)

Screen shot of the entirety of the graded homework question that includes:

- The question and answer text
- Points earned
- Grader comments (if any)

Dissection section that:

- Includes the correct answer in your own words/voice
- Explains clearly and in detail what about your answer was incorrect or incomplete
- Explains clearly and in detail what about the correct answer is different than your initial answer
- Explains clearly and in detail how it correctly or more fully answers the question



Can I use AI to help me understand a genetics concept?

Yes. You may use AI to explore, clarify, or review *general concepts* from lecture or readings, just as you would use a textbook or online resource.

Can I use AI while I am actively working on a homework question?

No. Once you begin working on a homework assignment, AI tools may not be used in connection with that assignment.

Can I paste a homework question into AI just to “check my understanding”?

No. Entering any homework prompt, data, figure, or scenario into an AI tool is not allowed, regardless of intent.

Can AI help me figure out how to solve a similar problem?

No. If the explanation would reasonably help you answer the homework question, then it is considered unauthorized assistance.

Can I use AI to reword, edit, or improve an answer I already wrote?

No. All homework must be written entirely in your own voice, without AI assistance at any stage of producing the response.

What if I didn’t copy anything directly from AI?

That’s expected. The issue is not copying text, but whether AI contributed directly to producing the homework answer. Using AI to learn *before* the assignment is allowed; using AI to help generate the assignment itself is not.

